LETROZOLE

Postmenopausal only (for cancer)

Bone loss

CHEMICAL NAME 4,4′-(1H-1,2,4-triazol-1-ylmethylene)dibenzonitrile
CLASS Non-steroidal aromatase inhibitor (hormonal anticancer)
TRADE NAMES Femara · Letroz · Letrozole (generic)

THERAPEUTIC USES / INDICATIONS

- Hormone receptor-positive breast cancer in postmenopausal women — adjuvant, extended adjuvant, metastatic, neoadjuvant.
- Ovulation induction in PCOS (off-label, first-line in many guidelines).

ADMINISTRATION

- **PO** once daily at the same time, with or without food.
- Ovulation induction: days 3–7 of cycle.

DOSAGES

- **Breast cancer:** 2.5 mg once daily (5 years or as advised).
- **Ovulation induction:** off-label; specialist protocol only.

PHARMACOKINETICS

- **Peak:** 1 h. Bioavailability ~100%.
- Metabolised by CYP3A4, 2A6.
- Excreted in urine. **t½:** ~2 days.

PHARMACODYNAMICS

- Inhibits **aromatase (CYP19)** → blocks peripheral conversion of androgens to oestrogens → ↓ oestrogen by >95% in postmenopausal women.
- In premenopausal women, ↓ oestrogen → ↑ FSH (ovulation induction).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Hot flushes, sweating.
- Arthralgia, myalgia, joint stiffness.
- Fatigue, headache, hypercholesterolaemia.

ADVERSE EFFECTS (SERIOUS)

- Osteoporosis and fractures.
- Cardiovascular events (small ↑).
- Carpal tunnel syndrome; trigger finger.
- Teratogenic.

CONTRAINDICATIONS & CAUTIONS

- Premenopausal endocrine status (for breast cancer, unless ovarian suppression).
- Pregnancy, breastfeeding.
- Severe hepatic impairment.

KEY INTERACTIONS

- **Tamoxifen:** ↓ letrozole levels.
- **Oestrogen-containing products:** antagonise effect.

NURSING MANAGEMENT

- Baseline bone density (DEXA) and every 1–2 years; calcium/vitamin D ± bisphosphonate.
- Monitor lipids, joint symptoms.
- Confirm menopausal status.
- Fertility: ultrasound follicle monitoring.

PATIENT EDUCATION

- Take daily; joint pains are common — exercise and analgesics help; tell your team.
- Weight-bearing exercise, calcium and vitamin D for bones.
- Report fractures, persistent joint pain, mood changes.

KEY POINTS

- Aromatase inhibitor; postmenopausal ER+ breast cancer.
- Bone loss, arthralgia.
- 2.5 mg daily.
- Also ovulation induction in PCOS.

SOURCES CHECKED [906] Letrozole 2.5 mg Film-coated tablets (UK SmPC) [907] Letrozole, oral (US label) [908] Letrozole, oral (US label, Bryant Ranch Prepack)
Full references p. 432–457